

Diagnosing Memory-Write Decisions in Conversational Assistants Through Specification-Based Replay

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Conversational assistants with persistent memory must complete required corrections without corrupting stored facts. A final-state score cannot tell a missing proposal from a proposed correction withheld at admission, or a tie between admission rules from a lack of opportunity to differ. We present a specification-based replay protocol that holds generated extractions and proposals fixed, varies only the admission rule, scores whole executed states against a frozen reference, and reports exposure, the outputs on which the rules can differ. Applied to 48 Korean test cases, 21 model configurations, and three repetitions, with three further diagnostic sources, replay separates three evaluation confounds that a final-state score conflates. Attribution: the 33-point correction gain of evidence checks over an agreement gate on Gemini 2.5 Flash is the release of held corrections, since a rule that releases every hold recovers the same corrections, and in 16 of 17 traced holds the corrected tuple was absent from the extracted fact lists although present in affirmed witnesses. Specification: of the 82 reference violations that the gate avoids on saved general-review proposals, 71 come from holds on appended facts that every rule treats identically, and 52 of these withhold a fact that the reference requires, written in a different form; the gate also removes 40 previously successful changes. Opportunity: ties between rules on 12 of 21 configurations with controlled-path proposals and 16 of 21 with review proposals reflect zero exposure. The study began as a prespecified test of whether target-linked evidence improves completion over general review; it found no advantage, and replay shows why. Outcomes measure conformity to the reference, and the claims concern the existence and attribution of these confounds rather than their prevalence. Replaying a deterministic admission rule needs no new model call, so a rule can be checked on saved proposals before it is credited.

CCS Concepts: • **Computing methodologies** → **Discourse, dialogue and pragmatics**; *Nonmonotonic, default reasoning and belief revision*; *Intelligent agents*; • **General and reference** → Evaluation.

Additional Key Words and Phrases: conversational agents, persistent memory, memory correction, admission control, specification-based evaluation, failure diagnosis

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1 INTRODUCTION

Conversational assistants built on large language models increasingly keep a persistent memory of their users [11]. Updating this memory requires both completing requested corrections and preserving facts that should remain unchanged. Current systems decide whether to add, revise, or delete stored facts [4, 7, 23, 41, 42] and are often evaluated later through questions about long conversations [22, 39]. A downstream or final-state score summarizes an outcome, but by itself does not identify which proposed write, admission decision, or execution constraint produced it. This article studies that diagnostic gap under an explicit state specification.

A simple example illustrates two failures relative to a specification. Suppose that the stored opening year of a shop is 1991 and the reference requires a correction to 1993. The model can propose the required operation, yet the application keeps 1991 because an agreement check does not find the corrected tuple in the facts extracted from every reading of the utterance. Releasing the hold completes that correction. Releasing every eligible held correction can also admit an operation that replaces the participant's workplace with their son's workplace, violating the specified target.

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The evaluation distinguishes a withheld required change from an admitted reference violation and traces the decision responsible for each.

We therefore separate three stages: generation produces a proposal, admission decides which proposed operations may execute, and execution changes the persistent state. When generation is held fixed, every difference in the resulting state can be assigned to admission within the tested pipeline. Four rules then act on identical operations. AGREE admits a correction only when its tuple is shared by the extracted facts of all readings, and ALLOW admits every correction that AGREE holds for lack of shared support. LITERAL additionally requires the target record to be consistent and the new value to occur in every reading, and WITNESS further requires, for each reading, an affirmed quotation linked to the same target. AGREE and ALLOW bound the comparison, and the two evidence checks measure correction completion and reference violations within these bounds. ALLOW is not an admit-all baseline: it keeps every other hold, including appended facts without shared support. Because the rules can differ only on outputs that contain a held correction, we also count these outputs and call this count *exposure*. It distinguishes equal outcomes on exposed proposals from equal outcomes when no proposal reaches the rules' decision boundary.

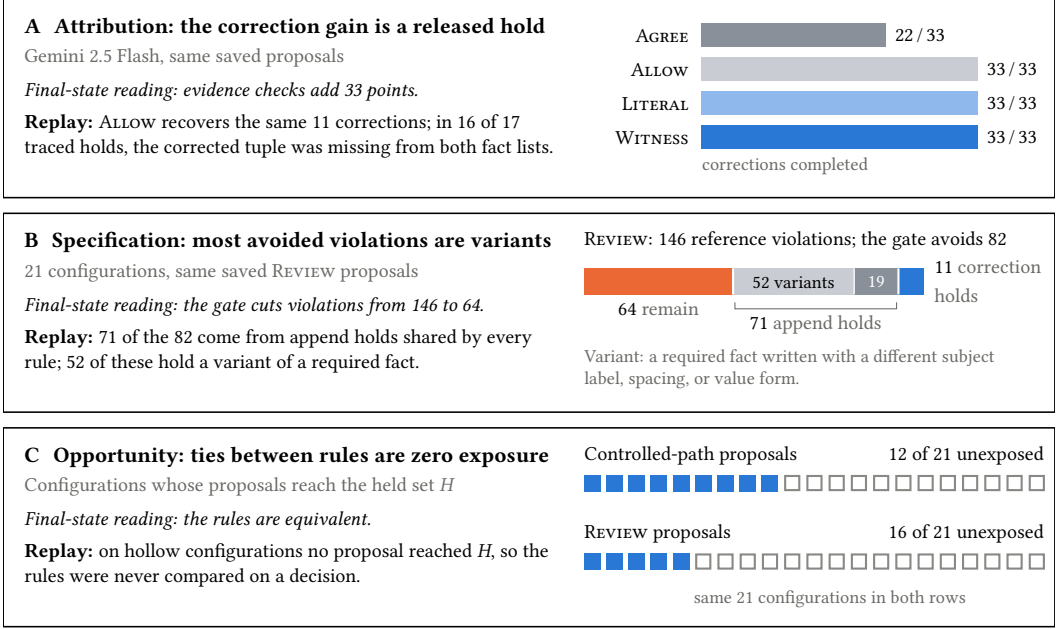
The study did not begin as a diagnosis. Its prespecified test asked whether admitting corrections only with target-linked evidence completes more required changes than general review; it did not (Section 6.5). A planned secondary comparison on the same saved artifacts showed a 33-point correction gain on one configuration, and replay with further controls traced that gain, and most violation reductions on saved review proposals, to decisions other than the evidence check. This article reports the protocol behind that explanation of a null result.

Three research questions organize the study. RQ1 asks where admission can intervene and which recorded conditions accompany withheld required corrections (Section 6.1). RQ2 asks how evidence requirements change correction completion and reference violations on fixed proposals (Sections 6.2–6.4). RQ3 asks how these effects compare with complete-pipeline outcomes and what changes when the gate is applied to saved general-review proposals (Sections 6.5 and 6.6).

This article makes three contributions:

- *A specification-based replay protocol for memory corrections.* The protocol extends fixed-candidate replay [17] to edits of identified records under competing readings, evaluates whole executed states, and reports whether proposals reach the boundary on which admission rules differ (Sections 3–5).
- *Three evaluation confounds for memory-write admission.* Each is a pair of situations that a final-state score cannot distinguish. *Attribution:* a difference credited to the admission rule arises in another stage; in 16 of 17 traced holds, the corrected tuple was absent from the extracted fact lists although present in affirmed witnesses, so the evidence checks' correction gain is the release of an extraction-induced hold. *Specification:* an avoided reference violation reflects how the reference matches written forms; of the 82 avoided on saved review proposals, 71 come from append holds shared by every rule, and 52 of these withhold a variant of a required fact. *Opportunity:* a tie reflects that no proposal reached the rules' decision boundary H , as on 12 of 21 configurations with controlled-path proposals and 16 of 21 with review proposals (Figure 1; Sections 6.1–6.6).
- *A measurement of admission effects and completion costs.* Complete-pipeline comparisons and replays of saved review proposals separate generation from admission and append holds from released corrections, reporting reference violations, successful changes, and invalid outputs together (Sections 6.5 and 6.6).

Scope of claims. The claims are existence and attribution claims: each confound is established by traced artifacts whose counts are exact and reproducible by replay, and one traced instance suffices



Counts pool three repetitions. Reference violations follow the frozen state specification.

Fig. 1. Three evaluation confounds that a final-state score cannot distinguish: attribution (A), specification (B), and opportunity (C). Each row sets what the final-state score shows beside what replay of the same saved artifacts shows. A: Gemini 2.5 Flash, strict corrections completed out of 33 observations under each rule. B: The 146 reference-violating states under REVIEW on the 21-configuration panel, split by whether the gate avoids them and, if so, by the held operation; “variants” are append holds whose tuple differs from a required fact only in subject label, spacing, or value form (Section 6.6). C: The 21 panel configurations, with controlled-path and with REVIEW proposals; hollow squares mark configurations on which no proposal reached the held set H .

to show that a final-state score can conflate the two situations. How often each confound arises in other systems is a prevalence question that these sources do not answer (Section 8). Only the complete-pipeline comparison was a prespecified test (Table 5), and outcomes measure conformity to the reference, not semantic safety or a validated deployment policy.

Sections 2–5 cover related work, the task, the rules, and the evaluation design; Sections 6–8 present the results, their implications, and the scope of the evidence.

2 RELATED WORK

2.1 Memory Updates and Admission in LLM Agents

Several systems give LLM agents a persistent memory that is updated through explicit operations. MemGPT treats the context window as fast memory and moves information to and from external storage through function calls [23]. Mem0 extracts candidate facts and applies one of four operations to existing memories: add, update, delete, or no operation [7]. A-Mem stores memories as linked notes whose attributes are revised as new notes arrive [41], and MemoryBank, recursive summarization, SeCom, LD-Agent, and Generative Agents maintain or select conversational history in related ways [16, 24, 25, 37, 51]. Closest to the correction decision studied here, Bae et al. cast memory management in Korean long-term conversation as pairwise operations between

a stored sentence and a new session summary (pass, replace, append, or delete) [4]. They compared trained classifiers with natural language inference models mapped to these operations and evaluated operation accuracy, set-level overlap, and human ratings of later sessions. Memory-R1 learns add, update, delete, and no-operation decisions from a reward based on downstream answer correctness [42], and Zep invalidates, rather than deletes, knowledge-graph edges that newer information contradicts [28].

A recent line of work treats the write itself as an admission decision. A-MAC learns an admission policy from interpretable factors (future utility, factual confidence, semantic novelty, temporal recency, and a content-type prior) and evaluates admission decisions against labeled candidate memories [47]; ConsistencyGate samples the same model several times to score whether a candidate fact is supported by its source and admits the fact above a threshold [49]; and DerivAudit audits whether a stored memory follows from the history available when it was written, finding that support often lies beyond the writer’s citations and that verifiers still admit many unsupported memories [17]. STALE evaluates whether agents notice that a stored memory has been implicitly invalidated by later information [5]. Memory writes are also an attack surface: Agent-Poison and MINJA show that poisoned or injected records, the latter inserted through ordinary queries, later steer agent behavior [6, 8]. In our setting, the surrounding history is supplied rather than retrieved, so retrieval failure is not confounded with admission failure; how the position of information affects its use is a separate question [18].

2.2 State Revision and Competing Hypotheses

Dialogue-state tracking already models state revision. SOM-DST predicts an operation for each slot before generating values [15], and its oracle analysis attributes most errors to the operation predictor rather than to value generation, an early stage-level diagnosis of the kind we apply to memory writes. AG-DST amends an initially generated state [35], dataflow synthesis supports executable dialogue programs and non-destructive revision [3], and PoA generates semantic-action code that updates the dialogue state [33]. Other trackers select the relevant turns, suppress incorrect values, or rewrite queries to resolve coreference [12, 19, 36]. The second Dialog State Tracking Challenge supplied competing speech-recognition hypotheses for every turn and scored the tracked state turn by turn [13]; its setting is close to ours, although there the tracker produces the state itself. MultiWOZ 2.2 removes values that the system offered before the user accepted them [46], and MultiWOZ 2.4 removes values that only the system mentioned [44]; both revisions show why the timing and source of evidence matter. MultiWOZ 2.2 also lists the surface variants of a value as equally correct [46], a reminder that an exact-match reference can count an equivalent mention as an error (Section 6.6 returns to this issue). Our comparison intervenes after extraction and proposal are fixed: it evaluates the permission to execute a proposal and thus complements these state-generation methods.

2.3 Evidence, Clarification, and Evaluation

Clarification has long been modeled as a decision under uncertainty, from POMDP dialogue policies to question selection based on the value of information or the expected information gain [9, 10, 26, 45, 48]. Human clarification behavior, however, is only weakly tied to annotated ambiguity [21] and is not mirrored by model uncertainty [34]. Moreover, an uncertain detail can be unnecessary for the next action yet consequential for a durable write.

Disagreeing readings resemble the inter-context conflicts studied between retrieved sources [40]. Faithfulness to supplied evidence and satisfaction of the information need are distinct outcomes, and overlap-based faithfulness scores can be inflated by copying the evidence [1]. Likewise, a quoted span shows that a value occurs in the source, not that it supports the proposed fact; this

Table 1. Positioning the admission decision relative to memory management, admission control, memory auditing, and state tracking.

Prior work	Update decision	Evidence / evaluation
SOM-DST [15]	Predict slot operation, then value	Dialogue context; per-turn state accuracy
DSTC2 [13]	Track competing recognition hypotheses	ASR hypotheses; per-turn tracking accuracy
Mem0 [7]	Choose add, update, delete, or no-op	Retrieved memories; QA on LoCoMo
A-Mem [41]	Create, link, and revise notes	Linked-note context; long-conversation QA
Bae et al. [4]	Pass, replace, append, or delete	Trained and NLI classifiers; operation accuracy and human ratings
Memory-R1 [42]	Learn memory-management operations	Retrieved memories; answer-correctness reward
A-MAC [47]	Learned admission and conflict-aware updating	Utility and grounding factors; admission precision, recall, and F1
ConsistencyGate [49]	Admit a fact above a support threshold	Sampled support scores; contamination and admission precision/recall
DerivAudit [17]	Verify support for candidate memory writes	Fixed-candidate replay; supported retention and unsupported admission
STALE/CUPMem [5]	Detect stale beliefs and adjudicate updates	Implicit-conflict scenarios; state resolution and downstream behavior
LongMemEval [39]	Benchmark long-term memory	QA including knowledge updates and abstention
τ -bench [43]	Execute tools under domain policy	Policy and tool feedback; final database state
This work	Admit operations from a fixed proposal	Same extraction and proposal for every rule; recovery, reference violations, and exposure at the update turn

distinction motivates our comparison between literal evidence and target-linked witnesses. Attribution to identified sources can be defined and rated by people [27], entailment-based checkers are strong automatic detectors of factual inconsistency [14], and agreement across readings resembles self-consistency, which takes a majority over a model's own sampled outputs [38]; AGREE, in contrast, requires agreement across alternatives supplied with the input. τ -bench scores agents by the final database state after tool use [43], the kind of executed-state endpoint that we apply to a single turn. Self-Refine and Reflexion revise outputs at inference time [20, 31]; we use a two-call general review as the alternative complete pipeline and report a task-adapted Self-Refine cycle in the online appendix. Because the choice of instruction can itself change results [2], every condition receives the same task facts and retention constraints. Success varies predictably from one input to the next [29], and multi-turn agent evaluation increasingly separates component abilities such as action, memory, and planning [11]. Automatic and human assessments can agree [32, 50] or rank systems differently [30], so agreement has to be established for each endpoint; for this reason, we distinguish semantic support from agreement with the state reference. Table 1 positions the study.

Prior work already studies write-time admission and its costs. DerivAudit compares verification settings on the same candidate memories and measures both supported-memory retention and unsupported admission [17], so replaying fixed candidates is not in itself our contribution. What we add is the combination of replay with four features of the correction decision: the operation edits an identified existing record and must retire the obsolete value while keeping the identifier; the utterance arrives as several competing readings; outcomes are scored on the whole executed

state rather than per candidate; and exposure records whether the rules had an opportunity to differ. We do not compare directly with A-MAC or ConsistencyGate because their evaluations score the admission of candidate facts against labels (precision, recall, or contamination), whereas our unit is the whole executed state after a fixed proposal edits identified records. Either method could be added as a further rule on the same saved proposals, a comparison we leave to future work (Section 8).

3 TASK AND DECISION BOUNDARY

A test case is $x = (h, T, M, \pi)$, consisting of the preceding context h ; an indexed sequence $T = (t_1, \dots, t_k)$ of alternative readings of one current utterance, which may coincide; an initial memory state M ; and a case-level retention permission π . The readings are simultaneous alternatives for the same speaker's turn, not separate speakers or successive turns. A record has a persistent identifier and a fact tuple (subject, relation, value, time). The model proposes an action a , a reply u , and an ordered list of memory operations Q ; the application executes the admitted operations in one transaction and returns the state M' together with an execution log ρ .

Four operations are available: APPEND, CORRECT, HOLD, and no-write. A correction targets an existing record and replaces its value while keeping its identifier, whereas a hold leaves a proposed fact uncommitted. Retention permission is separate from factual support, because a clearly stated fact may still be disallowed for storage. Supplying the initial state lets the evaluation distinguish a new fact from an obsolete or protected one without having to infer it. For each case, the reference specifies the allowed actions, the required and forbidden tuples, the records that must remain unchanged, the obsolete versions to retire, and any records that must no longer appear; all of these requirements are checked on the executed state.

We assess required change and preservation together. On a positive case, at least one required fact is absent or an obsolete version must be retired, so holding every proposal fails. A correct target update can still fail whole-state conformity when it also appends a fact outside the requested edit. We score the action and persistent state against the reference and inspect reply-state discrepancies in selected traces; reply truthfulness is not a scored endpoint. In tables, "Viol." abbreviates reference violations. Table 2 defines the terms and endpoints used throughout.

4 CORRECTION ADMISSION RULES

4.1 One Proposal and Four Admission Decisions

The controlled path makes two model calls. The first returns an extraction F : for each reading, the atomic facts that the reading supports, and, for each pair of a stored record and a reading, one correction witness with a polarity label (affirmed, negated, uncertain, or absent), a new value, and a verbatim quotation. The second call receives the test case together with F , marked as fallible, and generates the final proposal. Every admission rule receives the identical test case, extraction, and proposal, together with an isolated copy of M , so changing the rule requires no further model call (Figure 2).

AGREE applies to every proposed APPEND and CORRECT operation. It admits an operation when the exact subject–relation–value–time tuple is supported by the fact list of every reading or is already present in M ; otherwise, it changes the operation to HOLD. AGREE was the conservative gate of the original protocol, fixed before collection, and is meant to keep a fact out of memory when the readings disagree about it. Shared support requires that the extraction validly cover all readings. Because the test compares tuples exactly as extracted, and because the extraction schema allows a reading's fact list to be empty, a correction can appear in the proposal and in the witnesses yet be missing from the intersection of the extracted fact lists. This is how the held set arises.

Table 2. Terms and endpoints. With three repetitions, the correction, positive, and all-case denominators are 33, 84, and 144 per configuration; with one repetition, they are 11, 28, and 48.

Term	Definition
Reading	One written alternative of the current utterance, supplied as text.
Extraction F	Reading-level fact lists and correction witnesses produced by the first model call.
Proposal (a, u, Q)	Action, reply, and memory operations produced by the second model call.
Held set H	CORRECT operations that AGREE holds only for lack of shared support, although the extraction validly covers every reading.
Exposure	Outputs on which the rules can decide differently, that is, outputs that contain an operation in H , reported with the number of distinct cases they represent.
Complete pipeline	A system that generates its own proposal. REVIEW: a draft plus one general-review call.
Controlled path	Extraction, proposal, and WITNESS admission; reply rendered from the execution log.
Strict correction completion	Allowed action and exactly the reference-specified whole state on the 11 correction cases.
Positive-change completion	Allowed action and exact whole state on the 28 positive cases (the prespecified primary endpoint).
Reference-violating state	Forbidden or extra write, or damage to a protected fact, as defined by the reference (all 48 cases).
Whole-state conformity	Executed state meeting every state requirement of the reference (all 48 cases).
Negative preservation	Preservation of the state on the 20 cases that require no change.

Table 3. Admission rules applied to the same proposal. Outside H , every rule keeps AGREE's decision; retention permission and execution checks are shared.

Rule	Admission within H
AGREE	Keeps the conservative hold when shared extracted support is missing.
ALLOW	Admits every otherwise eligible correction in H .
LITERAL	Checks the existing target and the unchanged subject, relation, and time; requires the new value in every reading and current-turn evidence.
WITNESS	Additionally requires one affirmed, matching, target-linked witness per reading, supported by an exact source quotation that contains the value.

Let $H(x, F, Q)$ be the set of operation indices j for which AGREE holds a proposed CORRECT only because its support is not shared, although every reading is validly covered. All differences between AGREE and the other three rules lie within H ; outside H , every rule keeps AGREE's decision (Table 3). ALLOW restores every correction in H and nothing else, so an APPEND that lacks shared support stays held under every rule. Two predicates define the evidence checks. For LITERAL ($L_j = 1$), the proposal must name one existing target; preserve its subject, relation, and time; change its value; and cite the current turn in both the fact and the operation. In addition, the replacement value must occur literally in every reading. For WITNESS ($W_j = 1$), the same target and source conditions must hold, and each reading must supply exactly one witness for the target record that is labeled affirmed, repeats the proposed subject, relation, time, and value, cites the current

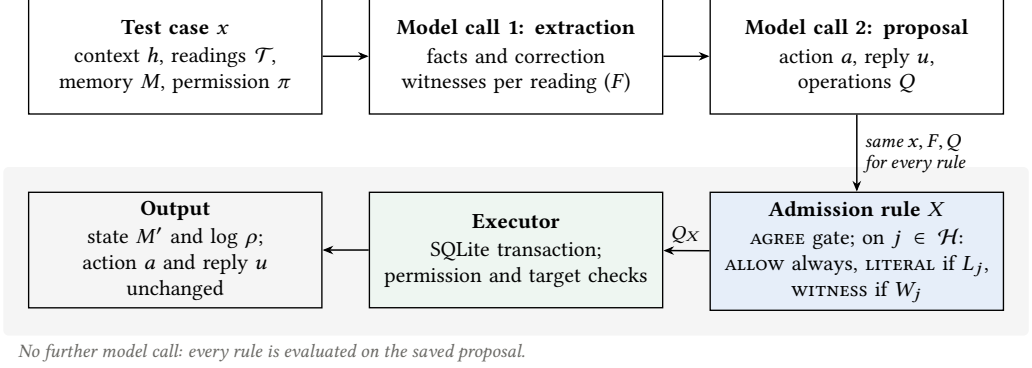


Fig. 2. Fixed-proposal architecture. Two generation calls produce one extraction and one proposal. Each admission rule is then applied to the same artifacts and executed on an isolated copy of the initial state.

turn, and quotes a verbatim span of that reading containing the replacement value. The application verifies the structure and linkage of these model-generated witnesses, but the polarity label itself is taken from the model.

Two examples show how the checks differ. The sentence “the shop opened in 1993, not 1991” satisfies both checks for an opening-year record. By contrast, “My son works at Daehan Post” satisfies the literal check for an update of the speaker’s own workplace but should fail the witness check, which requires every reading to affirm the value for the targeted record: the affirmed statement concerns the son.

4.2 Containment and the Opportunity to Intervene

The admitted sets of the four rules are nested. Write A_X for the indices in H that rule X admits; then

$$A_{\text{WITNESS}} = \{j \in H : W_j = 1\} \subseteq A_{\text{LITERAL}} = \{j \in H : L_j = 1\} \subseteq A_{\text{ALLOW}} = H. \quad (1)$$

The first inclusion holds because a passing witness carries the required target and source linkage and an exact quotation that contains the value, so the value occurs in every reading; the second follows from the restriction to H . This ordering concerns admitted operations only. It need not carry over to final-state accuracy when a transaction is rejected, when operations interact, or when a witness is wrong.

AGREE and ALLOW are therefore the two ends of this ordering rather than competing designs: no rule that acts on H can admit more held corrections than ALLOW or fewer than AGREE. We compare LITERAL and WITNESS with both ends: with ALLOW for the reference violations they prevent, and with AGREE for the corrections they recover and the reference violations they add. Because AGREE reads model-extracted facts, its holds can originate in extraction rather than in disagreement between readings; the saved extraction shows which of the two occurred (Section 6.1). WITNESS and LITERAL decide differently only on $K = \{j \in H : L_j = 1, W_j = 0\}$, where a hold may either protect the correct state or keep an obsolete fact unnecessarily. When H is empty, all four rules must agree. We therefore count exposed outputs and their distinct cases before interpreting any rule comparison.

Algorithm 1 Admission on a fixed proposal. Reply rendering is a separate downstream option.

Require: case $x = (h, T, M, \pi)$, extraction F , proposal (a, u, Q) , rule X (AGREE, ALLOW, LITERAL, or WITNESS)

Ensure: action a and reply u unchanged; executed state M' and log ρ

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1: if  $Q$  fails schema validation then
2:   return  $(a, u, M, \emptyset)$  ▷ no write; scored as an invalid output
3: end if
4:  $Q_X \leftarrow \text{AGREEGATE}(x, F, Q)$  ▷ holds APPEND and CORRECT without shared support
5: for all  $j \in H(x, F, Q)$  do
6:   if  $X = \text{ALLOW}$  or  $(X = \text{LITERAL and } L_j = 1)$  or  $(X = \text{WITNESS and } W_j = 1)$  then
7:      $Q_{X,j} \leftarrow Q_j$  ▷ restore the proposed correction
8:   end if
9: end for
10:  $(M', \rho) \leftarrow \text{EXECUTE}(M, Q_X, \pi)$ 
11: return  $(a, u, M', \rho)$ 

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4.3 Execution and Complete Pipelines

The application applies operations transactionally to a separate SQLite state for each rule and records read-after-write execution logs; the executor enforces retention permission and target existence for every rule. Because admission does not regenerate the action or the reply, differences in state outcomes between rules can be assigned to admission alone within this pipeline. For the same reason, a held write can coexist with a reply that asserts the held value (Online Appendix H), and the unchanged reply is not credited to admission. WITNESS uses the same two calls per output as AGREE, and ALLOW and LITERAL are computed from the saved proposals (Algorithm 1; Online Appendix C).

The comparison pipeline, REVIEW, generates its own proposal from a draft and one general-review call, so it and the controlled path use the same number of calls. The original protocol also ran a single direct draft and one draft–feedback–refine cycle adapted from Self-Refine [20]; they share REVIEW’s initial draft where the design permits and are reported in Online Appendices A and F. The prespecified primary test compares the controlled path, with its reply rendered from the execution log, with REVIEW. Rendering does not change any state endpoint, so the tables report the common state score under WITNESS (Online Appendix A).

5 EVALUATION DESIGN

5.1 Sources and Model Configurations

The main source contains 60 Korean test cases written by the authors with AI assistance: 12 development cases and 48 held-out cases spanning 12 problem families of four cases each. Each held-out case was run three times per model configuration. A private reference, fixed before the main collection, specifies the requirements listed in Section 3, including the distinction between replacing a value and removing its record. The collection script received the public cases but not the reference, and the model configurations, prompts, admission rules, and evaluator were fixed in a frozen protocol before collection. These measures kept the expected answers out of generation and execution. We use Korean because widely used memory benchmarks are in English [22, 39] and because Korean writing yields several surface forms of one fact (word spacing varies in practice, and a person may be named by a kinship term, a given name, or both), which stresses exact tuple matching (Section 6.6).

Table 4. Sources and their roles. Additional models reuse the main cases, and each source keeps its own denominators.

Source	Cases	Repetitions	Role
Main Korean cases	48	3	Fixed-proposal and complete-pipeline comparisons
Contrast set	24	1	12 supported and 12 unsupported edits
Boundary probe	12	1	Six paired edits; no exposure
Public dialogue probe	16	1	10 eligible self-repairs; six unscored

All alternative readings are written text. The main-source readings were authored with AI assistance, the contrast and boundary readings were generated as authored probes, and the public source adapts written task-dialogue turns. Within a case, the readings can be identical, differ in surface form, or differ in value, person, or polarity. They play the role of competing recognition hypotheses but were not produced by a speech recognizer; how often they reach the held set therefore characterizes these sources rather than the error pattern of any recognizer. Figure 3 lists the 12 problem families, which range from explicit corrections and self-repairs to conflicting transcripts, equivalent variants, and wrong subjects.

The original protocol planned a Holm-adjusted primary comparison for five model families. Three were collected—Gemini 2.5 Flash, Llama 3.1 8B, and Qwen3 14B (local 4-bit)—and two were not (Online Appendix C). Qwen3 14B results are reported in full, and the original five-slot Holm accounting is unchanged (Online Appendix C). Six later Gemini configurations (2.5 Pro, 2.5 Flash-Lite, 3 Flash Preview, 3.1 Pro Preview, 3.5 Flash-Lite, and 3.8 Flash) reuse the 48 cases and were added after the earlier results had been inspected; together with Gemini 2.5 Flash, they form the seven Gemini configurations of the four-rule comparison on identical proposals. Qwen3 14B adds a separate saved-proposal comparison of the literal check.

The model panel comprises 21 configurations that completed all 48 held-out cases in all three repetitions and produced at least 95% valid final outputs in both REVIEW and the controlled path (at most seven invalid outputs of 144 in either). This validity criterion was set after collection; it applies to complete pipelines rather than to semantic outcomes, and it keeps every observation of a selected configuration, including residual invalid outputs. The panel includes seven Gemini, eight Claude, and three GPT configurations together with DeepSeek 3.2, GLM 5, and Qwen3 14B, for 3,024 case-repetition episodes and 21,168 saved condition-score rows. Of the 28 configurations with complete triplets, seven fall below the criterion, among them Llama 3.1 8B, whose controlled path produced 76 valid outputs of 144; its results are reported in Section 6.5 and Online Appendix B. Model labels denote the recorded provider or runtime configurations and do not imply matched parameter counts, precision, or reasoning budgets.

Table 4 lists three follow-on sources. The terms “supported” and “unsupported” for their edit strata denote the source-specific reference labels; they are not independent judgments of every generated output. The 24-case contrast set was written after the initial recovery finding by a separate source-writing agent that received the schema but neither the rule code nor earlier outputs. The 12-case boundary probe pairs six edits with six non-edits. The public source uses 16 selected user turns and prior task-state subsets from MultiWOZ 2.4 [44]; its reference classifies 10 cases as same-task self-repairs, and the six plan-change or ambiguous cases remain in the archive without automatic correctness labels.

5.2 Endpoints

The primary endpoint prespecified in the original study is positive-change completion: joint success of the action and the whole state on the 28 of the 48 main cases that require a new fact or the retirement of an obsolete version. A successful output must be valid, choose an allowed action, contain the required facts, omit forbidden and obsolete tuples, preserve protected facts, satisfy the required removals, and introduce no unlisted fact. Provider failures and invalid final outputs stay in the denominator. Strict correction completion applies the same criterion to the 11 correction-eligible cases, so an out-of-scope addition fails it even when the target value is right. The reference-violation endpoint flags forbidden or extra writes and damage to protected facts across all 48 cases; reference violations and incompleteness can overlap. Negative preservation uses the remaining 20 cases. The contrast set scores exact, scope-compliant state on its two strata of 12 cases, and the public probe reports agreement on the focal part of the task state for its 10 eligible cases (Online Appendix F). Successful positive changes and reference-violating states are always reported together, so that withholding more operations is not mistaken for improving the task.

The reference defines the measured outcome, so the scores measure conformity to this specification: equivalent surface forms can fail exact matching, selected traces explain individual mismatches, and semantic support and reply-state consistency are not aggregate endpoints (Section 8).

5.3 Paired Analysis

For configuration m and case i , the paired difference is $d_{mi} = \frac{1}{3} \sum_{r=1}^3 [Y_{mir}(\text{ctrl}) - Y_{mir}(\text{REVIEW})]$, where Y is the endpoint indicator for repetition r and ctrl denotes the controlled path; the effect estimate averages d_{mi} over cases. Uncertainty is reported as 95% percentile intervals from 5,000 case-cluster bootstrap draws. The case is the resampling unit (48 clusters for all-case endpoints, 28 for positive completion, and 11 for correction completion), and the three repetitions are kept together within each cluster; we call these intervals case-cluster intervals. Archived intervals are reused only when their configuration, contrast, endpoint, and observation set match the reported estimate; otherwise, the point estimate is reported alone. The prespecified exact two-sided McNemar test, by contrast, counts a case as a success when it succeeds in at least two of three repetitions. It therefore tests a binarized case-level outcome, whereas the interval describes the mean difference; we report both for the primary comparisons. The original plan adjusted the five planned model-family tests with Holm's method (Online Appendix A).

Each completed original-protocol run made $48 \times 3 \times 6 = 864$ physical calls across the shared generation stages, of which each two-call pipeline (REVIEW or the controlled path) used 288; no failed call was retried within that run, and later collection phases keep their own provenance records. Gemini 2.5 Flash used temperature 0.2, disabled thinking, and 2,048 output tokens (512 for feedback). Later configurations used their supported thinking settings and larger output caps, and the 3.5 Flash-Lite requests left the temperature at the provider default; the panel therefore compares configurations as run rather than at equal compute (Online Appendix C). Configuration-level direction counts tally observed directions; they are not tests of wins or estimates over a population of models.

5.4 Confirmatory and Exploratory Analyses

Only the comparison of the controlled path with REVIEW was prespecified as a test. Several comparisons on identical proposals were added after earlier results had been inspected; Table 5 records the status of each analysis, and Section 6 refers to it rather than repeating it.

Table 5. Status of the reported analyses. “Prespecified” does not imply public preregistration, and additional model coverage does not turn an analysis into a confirmatory one.

Analysis	Inferential role
Controlled path versus REVIEW (original families)	Prespecified primary test; five-slot Holm accounting retained.
AGREE versus WITNESS	Planned secondary contrast; descriptive.
ALLOW and LITERAL	Post hoc controls on saved proposals.
Hold traces; exposure on all 21 configurations	Post hoc, from saved extractions and gate logs.
Variant audit of avoided violations	Post hoc mechanical comparison.
Six later Gemini configurations; 21-configuration panel	Exploratory reuse of the same 48 cases; panel criterion set after collection.
Contrast, boundary, and public sources	Follow-on diagnostic and transfer studies.
Normalization controls; REVIEW+LITERAL and +WITNESS	Retrospective; policies fixed before the new states were executed.
REVIEW+AGREE and +ALLOW;	Retrospective bounds added after the hybrid results
APPEND/CORRECT counts	were inspected; original rule definitions.
Family, case, and replay-time analyses	Descriptive.

Four additional policies were fixed before any new state was executed and were applied to the same 21 configurations, 48 cases, and three repetitions: normalized AGREE, normalized LITERAL, REVIEW+LITERAL, and REVIEW+WITNESS. Normalization applies Unicode NFC, case folding, and whitespace collapsing to subject and value equality in AGREE and to value containment in LITERAL; identifier, relation, time, source, and permission checks remain exact, and no output is rewritten. Each hybrid pairs the saved REVIEW proposal with the extraction from the same configuration, case, and repetition; an invalid extraction leaves the initial state and counts as an invalid output. All new conditions were executed with the frozen executor and scored with the archived evaluator and reference; as a fidelity check, normalized AGREE changed no gate decision and reproduces the saved AGREE scores on all 3,024 observations. After these hybrids had been inspected, REVIEW+AGREE and REVIEW+ALLOW were added as the two bounds of Section 4.2, with each APPEND and CORRECT decision logged; their 6,048 outputs were scored once without model calls. Pooled intervals resample all configurations and repetitions within a case together (5,000 draws, fixed seed).

6 RESULTS

We report component and complete-pipeline comparisons to separate two sources of outcome differences. Component replay varies admission while holding the extraction and proposal fixed. Complete-pipeline comparisons also allow generated artifacts to change.

6.1 Withheld Corrections Originate in Extraction

On Gemini 2.5 Flash, the agreement gate withheld 11 required corrections that the other three rules admitted. Changing admission from AGREE to LITERAL or WITNESS increases completed corrections from 22 to 33 of 33 observations, while reference-violating states remain at 3 of 144 under every rule (Table 6). The 11 recovered corrections come from five of the 11 eligible cases, and eight are repetitions of self-repair cases (Figure 3). The paired case-mean gain is 33.3 percentage points (95% case-cluster interval 12.1 to 57.6), and the case-level McNemar test, with four discordant cases all favoring WITNESS, gives $p = .125$. Positive-change completion rises from 69 to 80 of 84, and whole-state conformity rises from 128 to 139 of 144. Because ALLOW recovers the same corrections, releasing the hold is sufficient for recovery on these proposals.

Table 6. Comparisons on identical proposals in the seven Gemini configurations, in the order AGREE / ALLOW / LITERAL / WITNESS. Strict correction uses 33 observations from 11 cases, and the reference-violation endpoint uses 144 observations from 48 cases. The *H* column reports exposed outputs and, in parentheses, distinct cases. All counts combine three repetitions.

Configuration	Correction /33	Viol. /144	<i>H</i> outputs (cases)
Gemini 2.5 Flash	22 / 33 / 33 / 33	3 / 3 / 3 / 3	11 (5)
Gemini 2.5 Pro	31 / 31 / 31 / 31	3 / 9 / 4 / 3	6 (3)
Gemini 2.5 Flash-Lite	31 / 31 / 31 / 31	16 / 19 / 16 / 16	3 (1)
Gemini 3 Flash Preview	33 / 33 / 33 / 33	12 / 12 / 12 / 12	0 (0)
Gemini 3.1 Pro Preview	33 / 33 / 33 / 33	2 / 2 / 2 / 2	0 (0)
Gemini 3.5 Flash-Lite	32 / 32 / 32 / 32	7 / 12 / 7 / 7	5 (2)
Gemini 3.8 Flash	33 / 33 / 33 / 33	1 / 1 / 1 / 1	0 (0)

The opening-year correction illustrates this pattern (Online Appendix H). Both readings correct the year, and the final proposal already requests CORRECT. AGREE holds the operation because the corrected tuple is not shared by the facts extracted from every reading. WITNESS finds an affirmed, target-linked quotation in each reading and admits the update, and the execution log confirms that the stored year changed to 1993 while the record kept its identifier.

The saved extractions show the same pattern in all 11 holds. The held observations comprise all three repetitions of CT45 and CT46, two of CT01 and CT48, and one of CT42 (Online Appendix A). In each, the first call returned an empty fact list for both readings and recorded the correction only as two affirmed witnesses, one per reading, whose tuples match the proposal exactly. No hold involved a spelling, spacing, subject, or time mismatch, because no reading-level fact existed that could mismatch. In the 22 eligible observations that AGREE admitted, which include the remaining repetitions of CT01, CT42, and CT48, the corrected tuple appeared among the facts of both readings. Every hold therefore follows from the absence of the corrected tuple from both fact lists, not from the strictness of the tuple comparison, and string normalization before the agreement test could not have recovered these updates. Why the extractor placed these corrections only in the witness field remains open: the extraction prompt asks for the facts that each reading supports and does not direct corrections to the witness field.

The holds did not protect against disagreeing readings, the situation that AGREE is designed for: in every held observation, both readings carried an affirmed witness for the same corrected value. This is the attribution confound: the final-state comparison of AGREE and WITNESS, planned before collection, cannot tell whether the 33-point gain comes from target-linked evidence or from releasing these holds. The ALLOW control shows that any rule releasing these holds produces the same gain, and the saved extractions point to the extractor, not the admission rule, as the component to change; an extraction step that places corrections among the facts can be evaluated with the same protocol.

6.2 Literal Evidence Retains Completion While Reducing Reference Violations

Across the seven Gemini configurations, a single literal check accounts for most of the reference violations that the evidence checks prevent. Relative to ALLOW, WITNESS avoids 14 reference-violating state observations, and LITERAL avoids 13 of them (Table 7); these differences arise from three distinct cases reused across configurations and repetitions. Both checks keep ALLOW's strict-correction count on every configuration, so this protection costs no recovery. On Gemini 2.5 Pro, ALLOW produces 9 reference-violating states out of 144, LITERAL 4, and WITNESS 3, while strict

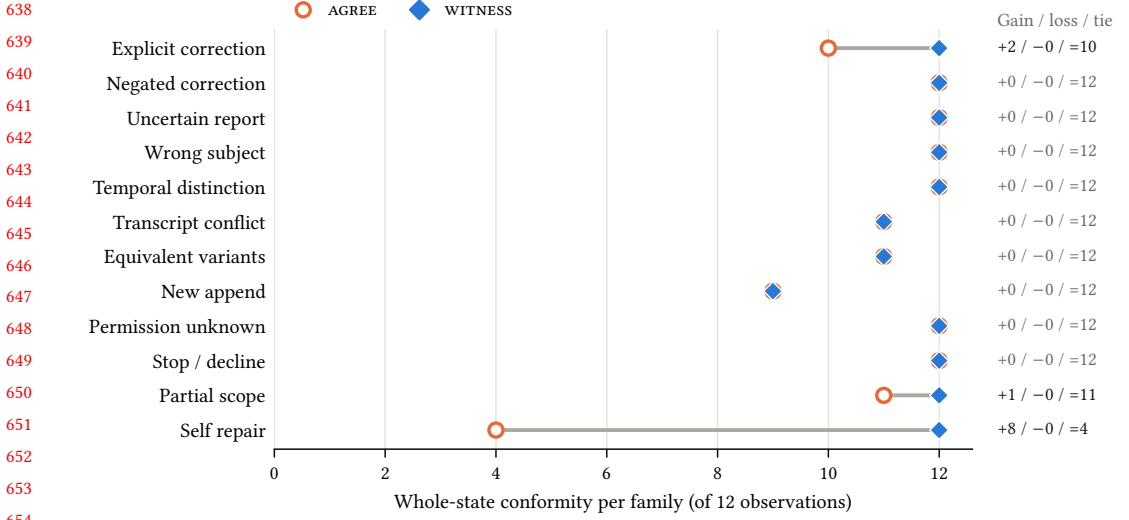


Fig. 3. Recovery by problem family on Gemini 2.5 Flash. The 11 observation-level gains come from five correction cases, and eight of them occur in self-repair; repetitions of a case are not independent cases.

Table 7. Reference-violating state observations prevented relative to ALLOW across three repetitions. The totals of 13 and 14 come from three distinct cases, not from 14 independent situations.

Configuration	LITERAL vs ALLOW	WITNESS vs ALLOW
Gemini 2.5 Flash	0	0
Gemini 2.5 Pro	5	6
Gemini 2.5 Flash-Lite	3	3
Gemini 3 Flash Preview	0	0
Gemini 3.1 Pro Preview	0	0
Gemini 3.5 Flash-Lite	5	5
Gemini 3.8 Flash	0	0
Gemini total	13	14

correction remains 31 of 33; the literal check also reduces reference violations from 19 to 16 on 2.5 Flash-Lite and from 12 to 7 on 3.5 Flash-Lite. Compared with the conservative end, WITNESS matches AGREE's reference-violation count on every configuration while recovering the 11 Flash corrections, and LITERAL recovers the same corrections while adding the single Pro error examined in Section 6.3.

Exposure shows where these rules have an opportunity to change a write. Gemini 3 Flash Preview, 3.1 Pro Preview, and 3.8 Flash produce no output that reaches H . Their ties are instances of the opportunity confound: no generated operation tested the difference between the rules, so the comparison supplies no admission effect on those proposals, and exposure should be measured before an admission effect is interpreted.

The saved gate decisions extend this count to all 21 panel configurations, whose controlled-path proposals were each scored under AGREE, LITERAL, and WITNESS (ALLOW only for Gemini and Qwen3 14B). Twelve configurations have no exposed output; the other nine have 49, from 12 cases (Online Appendix G). Both evidence checks release 12 of these holds, each completing

a required correction (11 on Gemini 2.5 Flash, one on DeepSeek 3.2); LITERAL alone releases two more, both reference violations (Gemini 2.5 Pro and GLM 5); and all three rules keep the other 35, all in cases that require no change, where the held state conforms to the reference. On most exposed outputs, the agreement gate thus withheld writes that the reference rejects, as designed; its holds of required corrections are concentrated in one configuration.

The saved Qwen3 14B proposals provide a separate configuration with substantial exposure. ALLOW produces 29 reference-violating states out of 144, whereas LITERAL and WITNESS each produce 20; the nine prevented observations come from three distinct cases, and strict correction completion remains 33 of 33. This result is consistent with the Gemini pattern for the literal check, and on this configuration the witness adds no further protection. Online Appendix H shows the three component changes side by side.

6.3 Target-Linked Evidence Blocked Person and Polarity Errors

The one reference-violating state that the literal check admits illustrates what the witness check adds. It occurs on Gemini 2.5 Pro (Figure 4). The utterance describes a son's workplace, and the same proposal both appends the son's workplace and corrects the participant's stored workplace. Because the replacement value occurs in both readings, LITERAL, like ALLOW, admits the second update. The witness for the participant's record is not affirmed, so WITNESS holds the update and keeps the participant's workplace. This single observation gives WITNESS one fewer reference-violating state than LITERAL (case-cluster interval -2.08 to 0.00 percentage points). Target-linked witnesses can thus block an update that literal occurrence cannot resolve: a value that appears in every reading but belongs to another person. The contrast set supplies a second such situation, a value that one reading negates (Section 6.4). The third is an uncertain report about a travel destination with a request not to change it (CT11): LITERAL admits the overwrite and WITNESS holds it, on GLM 5 controlled-path proposals (Section 6.2) and on saved Qwen3 14B review proposals (Section 6.6). The witness rule's added effect thus rests on four observations from three situations. Literal occurrence cannot tell whose fact a value is or whether a reading affirms it; binding the evidence to the target record can, and in these four observations it changed the executed state.

6.4 Additional Correction Sources

The 24-case contrast set, written after the initial finding (Table 5) and run once, shows the same recovery and the same immediate cause on new cases. Exact full-state success on the 12 supported edits is 5 for AGREE, 10 for WITNESS, LITERAL, and ALLOW, and 12 for REVIEW; preservation on the 12 unsupported edits is 9 for AGREE and WITNESS and 8 for LITERAL, ALLOW, and REVIEW. AGREE held six proposed corrections. For the five holds on supported edits, the saved extractions show the same pattern as on the main source: both readings had an empty fact list, and each had one affirmed witness matching the proposal. WITNESS, LITERAL, and ALLOW restored all five, so recovery again followed from releasing the hold. The sixth hold is the only one of the 17 traced holds (11 on the main source and six here) that reflects disagreement between the readings, the situation that AGREE is designed for. In this residence case (Online Appendix H), the first reading yields the new residence as a fact and an affirmed witness, whereas the second, which negates living in Changwon, yields no fact and a negated witness. Because the value occurs in both readings, LITERAL admits the overwrite, as ALLOW does, whereas WITNESS holds it; the shared reply nevertheless asserts Changwon even under WITNESS. All three releasing rules updated the intended target in the 12 supported cases, and two outputs also appended an out-of-scope fact, which shows that target recovery and whole-edit scope are distinct outcomes (Online Appendix D). In the 12-case boundary probe, no output reached *H*, and all rules achieved 4 of 6 supported edits and 5 of 6 preserved states, a tie with no opportunity to differ (Online Appendix E).

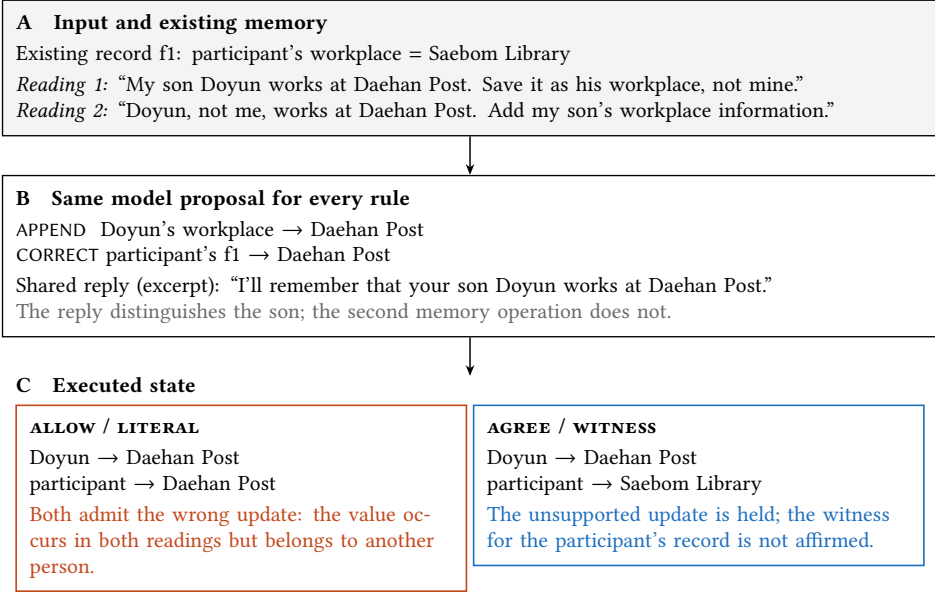


Fig. 4. Target binding changes the executed state. LITERAL accepts a workplace value that appears in both readings, whereas WITNESS rejects assigning the son's workplace to the participant. A correct-looking reply does not by itself guarantee a correct memory operation.

The public task-dialogue probe marks a limit of literal evidence. Among 10 eligible self-repairs from 16 selected MultiWOZ cases, agreement on the focal task state is 10 for REVIEW, 7 for ALLOW, 4 for LITERAL and WITNESS, and 3 for AGREE (Online Appendix F adds the two other original-protocol pipelines). Exact matching can hold values that differ only in case, such as "Thursday" and "thursday", and this is one plausible explanation for part of the gap. In the saved Thursday example, the extracted fact list is empty, so normalizing existing facts cannot repair the hold. A lexical diagnostic with Unicode NFC, case folding, and whitespace normalization does turn the value-to-source and value-to-quotation containment checks from false to true; this identifies a possible surface-matching repair for LITERAL and WITNESS, not a measured change in task success (Online Appendix F). On this external source, the evidence checks trail general review by six of the 10 eligible repairs, which makes surface matching the next component to test there.

6.5 Complete Pipelines Across Model Configurations

Across model configurations, the controlled path was not uniformly better than general review. Among the 21 panel configurations, positive-change completion increases on 9, decreases on 9, and ties on 3, and reference violations decrease on 12, increase on 7, and tie on 2 (Figure 5). Nine configurations improve on both outcomes and seven worsen on both; of the remaining five, two reduce reference violations only, one reduces both reference violations and completion, one reduces completion only, and one ties on both. Only one configuration therefore shows a simple trade-off between completion and reference violations. Because most configurations that change move both outcomes in the same direction, one plausible explanation is that the pipelines differ in the proposals they generate, not only in admission; Section 6.6 therefore applies the gate to saved review proposals.

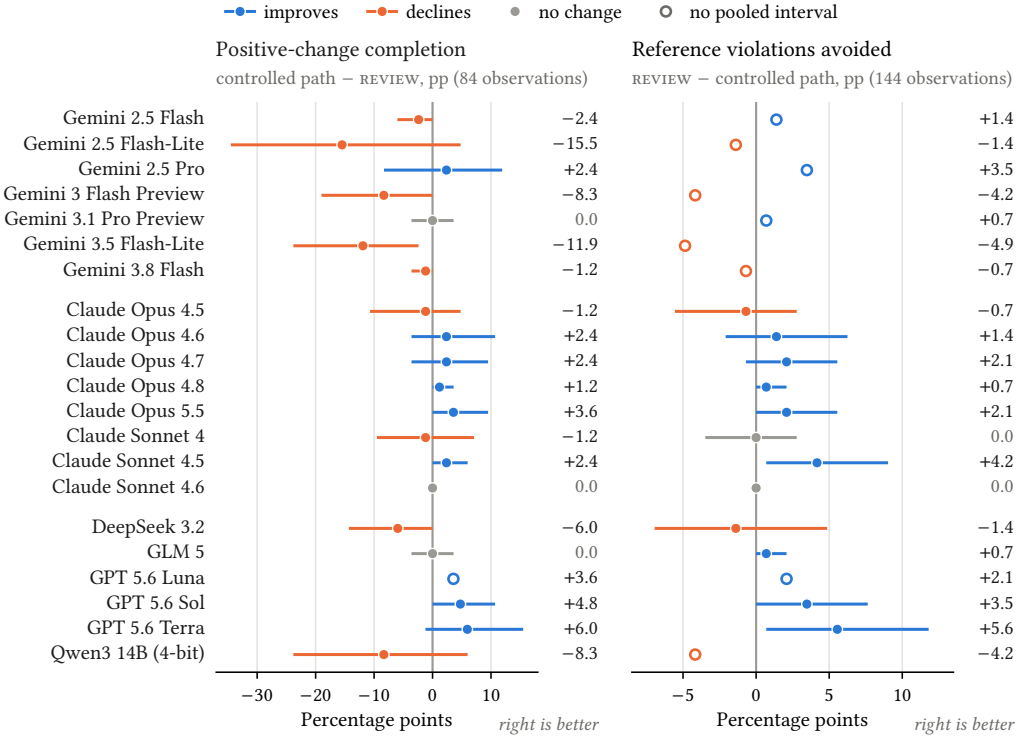


Fig. 5. Effects of the complete pipelines on the same 48 cases over three repetitions, oriented so that improvement is to the right (more positive completions and fewer reference-violating states). Positive completion uses 84 observations from 28 cases, and the reference-violation endpoint uses 144 observations from 48 cases. Blue marks improvement, orange decline, and gray no change; values at the right are point estimates in percentage points. Lines show the matching archived pointwise 95% case-cluster intervals, and open points mark configurations without a complete pooled interval. Configurations are not independent replications of the source.

Online Appendix G lists all 21 configurations. GPT 5.6 Terra, for example, completes 76 positive changes against 71 for REVIEW with 5 reference-violating states against 13, whereas Qwen3 14B completes 66 against 73 with 20 against 14, although its literal check improves the execution of fixed proposals (Section 6.2). Such cases motivate keeping the review proposal fixed while varying its admission.

The prespecified primary comparison, which concerns the original model families, found no completion advantage. On Gemini 2.5 Flash, REVIEW completes 82 of 84 positive-change observations and the controlled path 80 (difference -2.4 percentage points, 95% interval -6.0 to 0.0 ; McNemar $p = 1.00$); both recover all 33 strict corrections, and reference-violating states fall from 5 to 3. On Qwen3 14B, the difference is -8.3 points (-23.8 to 6.0). Llama 3.1 8B, which is outside the panel because its controlled path produced 68 invalid final outputs, completes 6 of 84 against 29 for REVIEW (-27.4 points, -44.0 to -10.7 ; Holm-adjusted $p = .0586$); its invalid outputs point to producing valid proposals, not admission, as its main limitation. Replay explains why the fixed-proposal gain does not carry over: the gain corrects the conservative admission decision, and general review itself completes all 33 strict corrections on Gemini 2.5 Flash. The direction counts

also depend on the validity criterion: without a threshold, across all 28 complete configurations, positive completion is higher on 10, lower on 15, and tied on 3, and reference violations are lower on 16, higher on 10, and tied on 2 (Online Appendix G).

The direction of the difference is also unstable across repetitions. Per repetition, positive completion is higher on 7, 6, and 7 configurations and lower on 8, 7, and 7, and 11 configurations change direction across repetitions (a tie counted as a direction); for this reason, all headline estimates pool the three repetitions by case. Strict correction completion under the controlled path matches REVIEW on 17 configurations and is lower on 4.

6.6 Most of the Gate's Effect Comes from Append Holds

Surface normalization does not account for the recovery gap: across the 3,024 panel observations, normalized AGREE and LITERAL leave every executed state unchanged (673 and 685 corrections; 131 and 133 reference-violating states), because the gap lies in facts absent from the extraction.

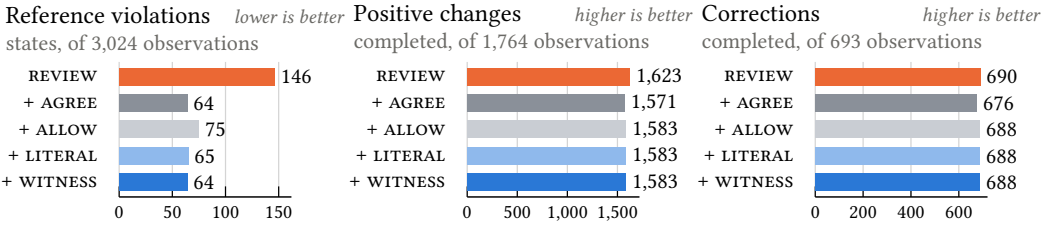
In this replay, applying the gate to saved REVIEW proposals more than halved the number of reference-violating states (Table 8, Figure 6). With the literal check, reference-violating states fall from 146 to 65; with the witness check, they fall to 64, and neither paired comparison introduces a new observation with a reference violation. REVIEW+WITNESS therefore prevents 82 reference-violating state observations across 16 distinct cases, a reduction of 2.71 percentage points over all 3,024 observations (95% case-cluster interval 1.39 to 4.30). Reference violations decrease on 17 configurations and tie on 4. Because the hybrid runs the full gate, AGREE holds every APPEND and CORRECT that lacks shared extracted support, and the evidence check decides only which held corrections return. The two bounds on the same proposals separate these parts: REVIEW+AGREE already reaches 64 reference-violating states but completes 676 corrections, whereas REVIEW+ALLOW completes 688 corrections and produces 75.

Counting the original decisions by operation locates the difference (Table 8; Online Appendix I). All four rules treat APPEND identically and hold 108 of 1,087 proposed appends; these holds account for 71 of the 82 prevented reference-violating states, from 13 cases, and for all 36 gate-related losses of positive changes. The other 11 prevented states involve corrections only. AGREE holds 23 of 762 proposed corrections, in outputs from eight cases and five configurations; on the other 16 configurations no REVIEW proposal reaches *H*. ALLOW releases all 23, LITERAL 13, and WITNESS 12. Each of the 12 releases shared by these rules completes a required correction: 11 are Gemini 2.5 Flash outputs from the five cases of Section 6.1, with the same repetition counts, because the hybrid reuses that configuration's saved extractions, and one is from DeepSeek 3.2 (CT46), the configuration of the single additional controlled-path recovery noted in Section 6.2. Each of the 11 corrections released only by ALLOW produces a reference-violating state; ten come from two conflicting-transcript cases that require no change, which LITERAL also holds, and one is the Qwen3 14B overwrite described below, which only WITNESS holds. Releasing these holds, as all three rules do, recovers 1.73 percentage points of strict correction relative to AGREE (95% case-cluster interval 0.58 to 3.17; five cases). The evidence checks differ from ALLOW only in the violations they avoid: 0.36 points for WITNESS (0.00 to 0.93; three cases), an interval that includes zero, and WITNESS differs from LITERAL by one observation (0.033 points, 0.00 to 0.10). Holding appended facts therefore changes the outcome far more than the choice of evidence check. No paired observation changed both operations, so each avoided violation is attributed to a single operation in these artifacts.

This reduction in reference violations has a measurable completion cost. The LITERAL and WITNESS hybrids, like REVIEW+ALLOW, lose 40 otherwise successful positive changes across nine cases and gain none (−2.27 percentage points, 95% interval −4.08 to −0.85); REVIEW+AGREE loses 52, adding the 12 corrections that the other rules release. Positive changes include required additions

Table 8. Gate applied to saved REVIEW proposals over the 21-configuration panel. Endpoint counts are case $\times$ repetition $\times$ configuration observations, not independent cases. The last column counts held decisions among 1,087 proposed APPEND and 762 proposed CORRECT operations, a different unit from the state counts; in each hybrid, two further decisions of each type were unavailable because the extraction was invalid.

Saved path	Viol. /3,024 ↓	Positive /1,764 ↑	Correction /693 ↑	Held APPEND / CORRECT
REVIEW	146	1,623	690	–
REVIEW + AGREE	64	1,571	676	108 / 23
REVIEW + ALLOW	75	1,583	688	108 / 0
REVIEW + LITERAL	65	1,583	688	108 / 10
REVIEW + WITNESS	64	1,583	688	108 / 11



21 configurations $\times$ 48 cases $\times$ 3 repetitions. Invalid final outputs: 21 under REVIEW, 29 under each admission rule.

Fig. 6. Admission applied to saved REVIEW proposals: the AGREE and ALLOW bounds and the two evidence checks. The reference-violation endpoint follows the fixed state reference, including exact tuple mismatches. The same 21 configurations and all three repetitions appear in every bar; denominators are 3,024 for reference violations, 1,764 for positive changes, and 693 for corrections. Online Appendix I reports every configuration with paired case-cluster intervals.

as well as corrections. The 40 losses have two sources. Thirty-six are APPEND holds on observations with a valid extraction, such as the required append described below. The other four, including both lost corrections (-0.29 points, -0.72 to 0.00), occur where the extraction was invalid and the hybrid left the initial state, which is also why invalid outputs rise from 21 to 29. Among observations valid in both pipelines, neither the LITERAL nor the WITNESS hybrid reduced strict-correction completion. Whole-state conformity moves from 2,849 to 2,810 with REVIEW+AGREE, 2,811 with REVIEW+ALLOW, 2,821 with REVIEW+LITERAL, and 2,822 with REVIEW+WITNESS: whole-state conformity falls even as reference violations fall. The cost is also unevenly distributed (Online Appendix I). On the four configurations where the gate prevents no reference-violating state (Claude Sonnet 4.6, Gemini 3 Flash Preview, 3.5 Flash-Lite, and 3.8 Flash), the hybrid loses 16 of the 40 positive changes, 14 through gate holds and 2 through invalid extractions. On Gemini 2.5 Flash-Lite and GPT 5.6 Terra, by contrast, it prevents 11 and 9 reference-violating states at a cost of 7 and 1 positive changes, the latter due to an invalid extraction. The completion-violation trade-off thus varies across configurations on the same cases.

Because the reference-violation endpoint follows the frozen exact state specification, not every prevented observation is a semantic error. Comparing each held tuple with the states that conform to the reference for the same case (Online Appendix I), 13 of the 82 prevented violations involve a forbidden tuple: the 11 held corrections and two Qwen3 14B appends of a disputed destination. The other 69 are unlisted appended facts, and 52 of them, from nine cases and 14 configurations, restate

a required fact with different spacing (6), subject label (29, e.g., “어머니순자” (“mother Sunja”) for “순자”), or value form (17, e.g., “자전거타기” (“riding a bicycle”) for “자전거”). The required fact is absent in these observations with or without the gate, so the hold lowers the violation count without changing completion; the mechanical comparison undercounts such variants. A substantive example is Qwen3 14B on CT11, repetition 3, which overwrites a destination despite a request not to change it; the Witness hybrid keeps the stored destination, whereas the Literal and Allow hybrids admit the overwrite. Conversely, on CT26, repetition 1, Claude Opus 4.5 correctly proposes to append that the participant learned from 오선생님 (“teacher Oh”), but Agree holds the append for lack of shared extracted support, and every hybrid withholds this required fact.

Applied after review, the gate thus acts mainly as an append policy: holding unsupported appends accounts for 71 of the 82 avoided violations and all 36 gate-related losses, while the choice of how to release held corrections changes the violation count by at most 11. Most of those 71 concern how a required fact is written rather than whether it should be stored, which is the specification confound; which appends to hold, and how their tuples are matched, are the decisions that most affect both endpoints here.

7 DISCUSSION

7.1 What Fixed-Proposal Evaluation Adds

Fixed-proposal evaluation connects executed-state scores to recorded admission decisions. Within an unchanged extraction, proposal, executor, and reference, it shows which required corrections are withheld and which admitted operations produce reference violations. Exposure identifies whether the proposed operations can distinguish the rules. Figure 1 sets the final-state reading of each confound beside its replay; the three rows draw on all three research questions. An AI-low control is necessary for these corrections but not sufficient. Existing evaluations report downstream answers or candidate-level admission precision and recall (Mem0 [7], A-MAC [47], and ConsistencyGate [49]; Table 1), a unit that does not show which held operation on a proposal changed the outcome. Running the control as a separately generated pipeline would also mix in generation variance: 11 of the 21 panel configurations changed the direction of the complete-pipeline difference across repetitions (Section 6.5). Holding the proposals fixed, as replay does, compares the rules on identical decisions.

7.2 Using the Diagnosis to Define Further Comparisons

Each diagnosis identifies a component to test (Table 9). The supplementary diagnostics in Section 7.4 implement three such comparisons: an extraction intervention, an LLM admission verifier, and a test with actual ASR alternatives. Their results distinguish the representation failure in the selected authored cases from disagreement across real recognition outputs. Each comparison retains completion and reference-violation endpoints, since stricter admission can lower both violations and completed changes.

Table 9. Comparisons motivated by the diagnoses, with completed exploratory tests distinguished from remaining work.

Observation	Comparison and current evidence
No proposal reaches H (opportunity)	Report zero opportunity; inspect generation or other stages when task outcomes remain poor.
Correct proposals held because extracted facts are absent (attribution)	The amended extractor removes four holds in five selected pairs (Section 7.4). Independent cases remain necessary.
Literal occurrence does not resolve person or polarity	Retain target-linked controls. The external ASR diagnostic also measures correct updates lost to admission (Section 7.4).
Exact matching penalizes equivalent values (specification)	Main-panel normalization leaves states unchanged (Section 6.6). Evaluate broader equivalence-aware references and append checks on new data.
A control changes local state scores	Compare complete pipelines and added inference cost. The saved-proposal LLM verifier supplies a separate exploratory comparison (Section 7.4).

7.3 Replay Cost and Pipeline Resources

Separating admission from generation allows deterministic policies to be compared on archived extractions and proposals without additional model calls. The public replay reconstructs gate decisions, transactional state changes, and frozen specification-based score fields from those artifacts. Replay resources are distinct from generation and serving latency. For context, generating the original Flash Review pipeline required 288 calls, 512,062 input and 54,918 output tokens, and 431.9 seconds of summed provider latency.

Deploying the gate, however, costs more than replaying it. The hybrid evaluated here combines the saved Review draft and proposal with the extraction of the controlled path, so a standalone implementation needs three generation stages rather than two. For Gemini 2.5 Flash, the summed stage times average 3.00 seconds for Review and 4.92 seconds for the hybrid, and the mean provider-reported token totals are about 3,965 and 5,991, respectively. These are sums of saved stages rather than serving latency. Because the extraction contains one witness for each pair of a stored record and a reading, larger memories would require scoping it to the records that a proposal targets.

7.4 Supplementary Mechanism and External-ASR Diagnostics

The following experiments were collected after the main analysis. They are exploratory, use separately frozen comparisons, and are not pooled with the 21-configuration, three-repeat panel. The anonymous artifact supplies their prompts, selection and scoring rules, recorded outcomes, paired intervals, and reproduction instructions.

Extraction intervention. We amended the extraction instruction to include explicitly affirmed replacement facts in the reading-level fact lists, even when correction witnesses were also present. The saved final proposals, admission rule, executor, and authored reference remained fixed. On five retrospectively selected Gemini 2.5 Flash correction cases, the newly collected

original instruction completed 1/5 corrections and the amended instruction completed 5/5: four Agree holds disappeared, with no reference-violating state in either arm. CT01 succeeded under both instructions. The intervention therefore recovers the remaining four holds in these pairs; it does not attribute the spontaneous recovery of CT01 to the amendment.

The complete 48-case Gemini 3.8 Flash comparison was unchanged: both instructions completed 11/11 corrections and achieved whole-state conformity in 47/48 cases. The initial Flash 2.5 collection yielded 15 paired cases before quota interruption; strict correction changed from 3/4 to 4/4 and whole-state conformity from 14/15 to 15/15, with paired intervals including zero. Completing the selected five pairs did not complete the planned full matrix. This intervention supports the missing-fact explanation for the selected failures, while its transfer to new cases remains open.

LLM admission verifier. Gemini 3.1 Pro approved or held each APPEND/CORRECT operation in 144 saved Review proposals: one selected repetition of 48 cases from each of three generators. The verifier left the original action and reply unchanged and was not fine-tuned. Table 10 compares it with Review and Review+Witness. All three conditions completed 11/11 corrections for each generator. The verifier matched the Witness results on Flash 2.5, recovered one positive change on Flash 3.8, and prevented two further reference violations on Qwen3 14B while retaining 24/28 positive changes. The key paired case-cluster intervals include zero. The verifier added 144 historical calls, with mean provider-reported stage time of 4.39 seconds; deterministic replay adds no model call.

Table 10. Exploratory verifier comparison on one repetition per generator. Each cell is Review / Review+Witness / LLM verifier. These are separate from the main panel.

Generator	Positive /28	Violations /48	Whole state /48
Gemini 2.5 Flash	27 / 27 / 27	2 / 1 / 1	46 / 47 / 47
Gemini 3.8 Flash	28 / 27 / 28	0 / 0 / 0	48 / 47 / 48
Qwen3 14B	24 / 24 / 24	4 / 3 / 1	42 / 43 / 44

External ASR alternatives. We selected one turn from each of 48 English restaurant dialogues in DSTC2 (Henderson, Thomson, and Williams, 2014), comprising 41 caller clusters and 433 retained, distinct ASR alternatives. The supplied previous human-derived goal state covered food, area, and price range; the current reference was withheld from generation. Selection was fixed before model outputs: 24 goal revisions, 12 denials without replacement, and 12 reference-enriched N-best cases. Original ranks were retained without likelihood weighting. This selected component test does not estimate population prevalence or full-benchmark tracking performance.

Gemini 3.1 Pro completed all 48 extraction/proposal pairs. On those same pairs, Witness reduced newly wrong writes from 3 to 0 but reduced complete-goal conformity from 33/48 to 18/48 (Table 11). The Witness-minus-Allow difference was -31.25 percentage points, with caller-cluster 95% interval [-45.83, -16.33]. Required-change completion fell from 23/36 to 6/36: -47.22 points [-62.86, -32.26]. Newly wrong writes changed by -6.25 points [-13.46, 0.00]. Of 20 held correction proposals, 17 were reference-correct and three were wrong. The focal authored-case signature of corrected facts absent from both readings was not observed: ten holds involved differing

extracted values and ten involved partial support or omission across actual alternatives. Literal admitted one negated food preference that Witness held, but Witness and Agree had identical aggregate outcomes.

Table 11. DSTC2 component diagnostic: 48 completed Gemini 3.1 Pro pairs, with 36 cases requiring a goal change. All rules replay the same proposals.

Rule	Required changes /36	Newly wrong writes /48	Complete goal /48
Allow	23	3	33
Agree	6	0	18
Literal	6	1	17
Witness	6	0	18

Gemini 3.8 Flash completed only 22/48 pairs before a quota error. Within that partial sample, Allow-to-Witness complete-goal conformity fell from 9/22 to 3/22 and newly wrong writes fell from one to zero. The two model samples remain separate; uncollected pairs are not zero scores. CareCall was not evaluated because the required original Korean data, human operation labels, and split linkage were unavailable.

Together, these diagnostics show why a successful local repair does not establish a generally better gate. Including corrected facts can remove selected extraction-induced holds, whereas agreement over actual ASR alternatives can reject correct changes. The verifier provides another admission rule with an explicit inference cost; its generator-dependent results do not establish uniform superiority.

DSTC2 source: Matthew Henderson, Blaise Thomson, and Jason D. Williams. 2014. *The Second Dialog State Tracking Challenge*. Proceedings of SIGDIAL, 263-272. [doi:10.3115/v1/W14-4337](https://doi.org/10.3115/v1/W14-4337). Source locators and hashes are provided in the artifact; restricted dialogue text is not redistributed.

8 SCOPE AND FUTURE WORK

The evidence is limited in several ways. It comes from 48 authored Korean cases, 11 of them correction-eligible, with written alternative readings, run on 21 model configurations with three repetitions; the contrast, boundary, and public sources extend it to further correction situations. Section 7.4 adds separately collected mechanism and external-ASR diagnostics. Configurations and repetitions broaden the range of generators while the cases stay fixed, so they add no independent situations. The rule differences also rest on few distinct cases: five for the re-covered corrections, three for the literal check's protection on controlled-path proposals and two on saved review proposals, and four observations from three situations for the witness-only holds. The controlled-path replays use our own two-call extraction and proposal pipeline; whether externally developed memory writers, such as Mem0's add, update, and delete proposals [7], produce extraction-induced holds is untested. Finally, the original primary comparison is the only confirmatory result, and it is null for completion; the diagnostic analyses, the later configurations, and the post hoc controls on saved proposals remain exploratory (Table 5).

The reference and witness checks constrain interpretation. Exact tuple matching penalizes equivalent expressions, as in at least 52 of the 82 violations avoided on review proposals, and no human judgment of semantic equivalence was collected; the polarity of a witness is a model output checked for structure and linkage rather than independently established entailment. The reference-violation endpoint is consequently a measure of conformance to the authored specification. It does not distinguish all semantic errors from representational mismatches, fewer violations need not imply better whole-state completion, and the counts supply no utility weighting between omitted and extra writes. The specification and evaluator are held fixed within each replay, so the comparisons remain interpretable within that boundary; the study makes no aggregate claim about user-perceived usefulness, semantic safety, or downstream conversational quality.

Further experiments should test whether these distinctions persist on independently authored cases and in other implementations. The extraction intervention in Section 7.4 addresses selected historical failures; a corrected extractor fixed on development data still requires an independent test. The DSTC2 diagnostic supplies actual ASR alternatives, but uses selected English three-slot dialogues with a supplied previous state. Its lost correct updates show that the authored-case mechanism cannot be assumed to explain external holds.

Append admission and semantic equivalence remain central open questions. Evidence checks for appended facts should be compared on the same saved proposals against references that accept equivalent forms or independent human judgments. The exploratory LLM verifier adds a model-based comparison; entailment and sampled-consistency verifiers [49], with thresholds fixed on development data, remain additional baselines. Serving latency, audio recognition accuracy, subsequent retrieval, and longitudinal interaction were not evaluated.

9 CONCLUSION

Specification-based replay makes memory-write outcomes traceable to admission decisions on fixed generated artifacts, and it separates three evaluation confounds that a final-state score cannot distinguish. Attribution: the correction gain of evidence checks over agreement is the release of holds that arose when extraction left the corrected tuple out of the fact lists, as in 16 of 17 initially traced holds. Specification: 52 of the 82 reference violations that the gate avoids after general review withhold a variant of a required fact, all within the 71 append holds shared by every rule, while the gate removes 40 previously successful changes. Opportunity: ties between rules on 12 of 21 controlled-path and 16 of 21 review-proposal configurations reflect zero exposure. Within held corrections, literal evidence supplied most of the protection against releasing wrong updates, and target binding changed the executed state in four observations: a value belonging to another person, a negated value, and, twice, a destination that the participant asked not to change.

The prespecified complete-pipeline comparison found no completion advantage over general review; replay explains why the fixed-proposal gain does not carry over, since general review already completes all 33 strict corrections on Gemini 2.5 Flash. For practice, the lesson is short: before crediting an admission rule, replay it on fixed proposals with an Allow bound (attribution), check that some proposal reached its decision boundary (opportunity), and audit avoided violations for variants of required facts while reporting completed changes alongside them (specification). Replaying the rules requires no new model call, yet it changes what a final-state comparison would conclude.

The supplementary intervention removes selected extraction-induced holds, while the actual-ASR diagnostic exposes a different cost: withholding reference-correct goal changes. These results strengthen the diagnostic use of fixed-proposal replay and delimit the transfer of a particular admission rule.

DATA, CODE, AND AI USE

An anonymized archive is available at <https://anonymous.4open.science/r/memory-admission-9E37/>. It contains the authored Korean cases and references, prompts, saved model outputs and failures, executed states, memory-write policies, evaluator, analysis code, and provenance hashes. Additional diagnostics are labeled separately. The DSTC2 supplement provides source locators, input hashes, reconstruction code, and transcript-free outcome flags; restricted source dialogue material is not redistributed. Offline replay requires no additional model calls.

The new experiments involved no human participants. OpenAI ChatGPT and Codex assisted with software development, test-case drafting, literature discovery, analysis code, and manuscript editing; the authors reviewed this content and take responsibility for it. English examples of the authored cases are translations of the Korean inputs that define those experiments.

Artifact condition codes are mapped to the paper in core/README: D = Direct, G = Review, SR1 = Refine, B = Agree, C = Witness, and EXACT_LITERAL = Literal. R and CR apply Agree and Witness with receipt-grounded reply rendering; the R_ prefix instead denotes saved-Review-proposal hybrids. The archived harmful_write field measures a reference violation, not independently judged semantic harm. The appendix retains its original experimental tables; additional diagnostic outcomes are reported in Section 7.4.

REFERENCES

- [1] Vaibhav Adlakha, Parishad BehnamGhader, Xing Han Lu, Nicholas Meade, and Siva Reddy. 2024. Evaluating Correctness and Faithfulness of Instruction-Following Models for Question Answering. *Transactions of the Association for Computational Linguistics* 12 (2024), 681–699. https://doi.org/10.1162/tacl_a_00667
- [2] Anirudh Ajith, Chris Pan, Mengzhou Xia, Ameet Deshpande, and Karthik Narasimhan. 2024. InstructEval: Systematic Evaluation of Instruction Selection Methods. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, 4336–4350. <https://doi.org/10.18653/v1/2024.findings-naacl.270>
- [3] Jacob Andreas, John Bufe, David Burkett, Charles Chen, Josh Clausman, Jean Crawford, Kate Crim, Jordan DeLoach, Leah Dorner, Jason Eisner, et al. 2020. Task-Oriented Dialogue as Dataflow Synthesis. *Transactions of the Association for Computational Linguistics* 8 (2020), 556–571. https://doi.org/10.1162/tacl_a_00333
- [4] Sanghwan Bae, Donghyun Kwak, Soyoung Kang, Min Young Lee, Sungdong Kim, Yuin Jeong, Hyeri Kim, Sang-Woo Lee, Woomyoung Park, and Nako Sung. 2022. Keep Me Updated! Memory Management in Long-term Conversations. In *Findings of the Association for Computational Linguistics: EMNLP 2022*. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 3769–3787. <https://doi.org/10.18653/v1/2022.findings-emnlp.276>
- [5] Hanxiang Chao, Yihan Bai, Rui Sheng, Tianle Li, and Yushi Sun. 2026. STALE: Can LLM Agents Know When Their Memories Are No Longer Valid? <https://doi.org/10.48550/arXiv.2605.06527> arXiv:2605.06527
- [6] Zhaorun Chen, Zhen Xiang, Chaowei Xiao, Dawn Song, and Bo Li. 2024. AgentPoison: Red-teaming LLM Agents via Poisoning Memory or Knowledge Bases. In *Advances in Neural Information Processing Systems*, Vol. 37. 130185–130213. <https://doi.org/10.52202/079017-4136>
- [7] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building ProductionReady AI Agents with Scalable Long-Term Memory. <https://doi.org/10.48550/arXiv.2504.19413> arXiv:2504.19413

- [8] Shen Dong, Shaochen Xu, Pengfei He, Yige Li, Jiliang Tang, Tianming Liu, Hui Liu, and Zhen Xiang. 2025. Memory Injection Attacks on LLM Agents via Query-Only Interaction. In *Advances in Neural Information Processing Systems*, Vol. 38. <https://openreview.net/forum?id=i7J62t2wtV>
- [9] Yijiang River Dong, Tiancheng Hu, Zheng Hui, Caiqi Zhang, Ivan Vulić, Andreea Bobu, and Nigel Collier. 2026. Value of Information: A Framework for Human-Agent Communication. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 42879–42896. <https://doi.org/10.18653/v1/2026.acl-long.1987>
- [10] Felix Gervits, Gordon Briggs, Antonio Roque, Genki A. Kadomatsu, Dean Thurston, Matthias Scheutz, and Matthew Marge. 2021. Decision-Theoretic Question Generation for Situated Reference Resolution: An Empirical Study and Computational Model. In *Proceedings of the 2021 International Conference on Multimodal Interaction*. ACM, 150–158. <https://doi.org/10.1145/3462244.3479925>
- [11] Shengyue Guan, Jindong Wang, Jiang Bian, Bin Zhu, Jian-Guang Lou, and Haoyi Xiong. 2026. Evaluating LLM-based Agents for Multi-turn Conversations: A Survey. *ACM Transactions on Intelligent Systems and Technology* 17, 4, Article 85 (2026), 40 pages. <https://doi.org/10.1145/3793671>
- [12] Jinyu Guo, Zhaokun Wang, Jingwen Pu, Wenhong Tian, Guiduo Duan, and Guangchun Luo. 2025. Multi-Perspective Dialogue Non-Quota Selection with loss monitoring for dialogue state tracking. *Expert Systems with Applications* 283 (2025), 127516. <https://doi.org/10.1016/j.eswa.2025.127516>
- [13] Matthew Henderson, Blaise Thomson, and Jason D. Williams. 2014. The Second Dialog State Tracking Challenge. In *Proceedings of the 15th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)*. Association for Computational Linguistics, Philadelphia, PA, USA, 263–272. <https://doi.org/10.3115/v1/W14-4337>
- [14] Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kuklansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. TRUE: Re-evaluating Factual Consistency Evaluation. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Seattle, United States, 3905–3920. <https://doi.org/10.18653/v1/2022.naacl-main.287>
- [15] Sungdong Kim, Sohee Yang, Gyuwan Kim, and Sang-Woo Lee. 2020. Efficient Dialogue State Tracking by Selectively Overwriting Memory. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 567–582. <https://doi.org/10.18653/v1/2020.acl-main.53>
- [16] Hao Li, Chenghao Yang, An Zhang, Yang Deng, Xiang Wang, and Tat-Seng Chua. 2025. Hello Again! LLM-powered Personalized Agent for Long-term Dialogue. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, 5259–5276. <https://doi.org/10.18653/v1/2025.naacl-long.272>
- [17] Hongjun Liu and Chen Zhao. 2026. Memory Is a Derivation: The Distributed-Evidence Paradox in Long-Term Agents. <https://doi.org/10.48550/arXiv.2609.36130> arXiv:2609.36130
- [18] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranajpe, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics* 12 (2024), 157–173. https://doi.org/10.1162/tacl_a_00638
- [19] Zhanghui Liu, Chaoxiong Zhou, Yuzhong Chen, and Zeping Zheng. 2026. Few-shot dialogue state tracking via context ranking and dynamic contrastive decoding. *Expert Systems with Applications* 300 (2026), 130197. <https://doi.org/10.1016/j.eswa.2025.130197>
- [20] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, Vol. 36. Curran Associates, Inc., 46534–46594. <https://doi.org/10.52202/075280-2019>
- [21] Chris Madge, Matthew Purver, and Massimo Poesio. 2025. Referential ambiguity and clarification requests: comparing human and LLM behaviour. In *Proceedings of the Eighth Workshop on Computational Models of Reference, Anaphora and Coreference*. Association for Computational Linguistics, 1–11. <https://doi.org/10.18653/v1/2025.crac-1.1>
- [22] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating Very Long-Term Conversational Memory of LLM Agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 13851–13870. <https://doi.org/10.18653/v1/2024.acl-long.747>

- [23] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. <https://doi.org/10.48550/arXiv.2310.08560> arXiv:2310.08560
- [24] Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Xufang Luo, Hao Cheng, Dongsheng Li, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Jianfeng Gao. 2025. SeCom: On Memory Construction and Retrieval for Personalized Conversational Agents. In *International Conference on Learning Representations*, Vol. 2025. 91851–91885. https://proceedings.iclr.cc/paper_files/paper/2025/hash/e56f394bbd4f0ec81393d767caa5a31b-Abstract-Conference.html
- [25] Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, 1–22. <https://doi.org/10.1145/3586183.3606763>
- [26] Sudha Rao and Hal Daumé III. 2018. Learning to Ask Good Questions: Ranking Clarification Questions using Neural Expected Value of Perfect Information. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2737–2746. <https://doi.org/10.18653/v1/P18-1255>
- [27] Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and David Reitter. 2023. Measuring Attribution in Natural Language Generation Models. *Computational Linguistics* 49, 4 (2023), 777–840. https://doi.org/10.1162/coli_a_00486
- [28] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. Zep: A Temporal Knowledge Graph Architecture for Agent Memory. <https://doi.org/10.48550/arXiv.2501.13956> arXiv:2501.13956
- [29] Wout Schellaert, Fernando Martínez-Plumed, and José Hernández-Orallo. 2025. Analysing the Predictability of Language Model Performance. *ACM Transactions on Intelligent Systems and Technology* 16, 2, Article 30 (2025), 26 pages. <https://doi.org/10.1145/3706118>
- [30] Ivan Sekulić, Mohammad Aliannejadi, and Fabio Crestani. 2024. Analysing Utterances in LLM-Based User Simulation for Conversational Search. *ACM Transactions on Intelligent Systems and Technology* 15, 3, Article 62 (2024), 22 pages. <https://doi.org/10.1145/3650041>
- [31] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems*, Vol. 36. Curran Associates, Inc., 8634–8652. <https://doi.org/10.52202/075280-0377>
- [32] Leila Tavakoli, Johanne R. Trippas, Hamed Zamani, Falk Scholer, and Mark Sanderson. 2025. Online and Offline Evaluation in Search Clarification. *ACM Transactions on Information Systems* 43, 1, Article 2 (2025), 30 pages. <https://doi.org/10.1145/3681786>
- [33] Dechuan Teng, Zheng Yan, and Wanxiang Che. 2026. Program of actions (PoA): Interactive semantic action parser as more powerful dialogue state tracker through code generation. *Expert Systems with Applications* 301 (2026), 130481. <https://doi.org/10.1016/j.eswa.2025.130481>
- [34] Alberto Testoni and Raquel Fernández. 2024. Asking the Right Question at the Right Time: Human and Model Uncertainty Guidance to Ask Clarification Questions. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 258–275. <https://doi.org/10.18653/v1/2024.eacl-long.16>
- [35] Xin Tian, Liankai Huang, Yingzhan Lin, Siqi Bao, Huang He, Yunyi Yang, Hua Wu, Fan Wang, and Shuqi Sun. 2021. Amendable Generation for Dialogue State Tracking. In *Proceedings of the 3rd Workshop on Natural Language Processing for Conversational AI*. Association for Computational Linguistics, 80–92. <https://doi.org/10.18653/v1/2021.nlp4convai1.8>
- [36] Bo-Hsiang Tseng, Shruti Bhargava, Jiarui Lu, Joel Ruben Antony Moniz, Dhivya Piraviperumal, Lin Li, and Hong Yu. 2021. CREAD: Combined Resolution of Ellipses and Anaphora in Dialogues. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, 3390–3406. <https://doi.org/10.18653/v1/2021.naacl-main.265>
- [37] Qingyue Wang, Yanhe Fu, Yanan Cao, Shuai Wang, Zhiliang Tian, and Liang Ding. 2025. Recursively summarizing enables long-term dialogue memory in large language models. *Neurocomputing* 639 (2025), 130193. <https://doi.org/10.1016/j.neucom.2025.130193>
- [38] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *The Eleventh International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=1PL1NIMMrw>

- [39] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2025. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. In International Conference on Learning Representations, Vol. 2025. 86809–86836. https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01Abstract-Conference.html
- [40] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge Conflicts for LLMs: A Survey. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Miami, Florida, USA, 8541–8565. <https://doi.org/10.18653/v1/2024.emnlpmain.486>
- [41] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. A-Mem: Agentic Memory for LLM Agents. In Advances in Neural Information Processing Systems, Vol. 38. Curran Associates, Inc., Red Hook, NY, USA, 17577–17604. <https://doi.org/10.52202/085713-0593>
- [42] Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Jinhe Bi, Kristian Kersting, Jeff Z. Pan, Hinrich Schütze, Volker Tresp, and Yunpu Ma. 2026. Memory-R1: Enhancing Large Language Model Agents to Manage and Utilize Memories via Reinforcement Learning. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, San Diego, California, United States, 12805–12825. <https://doi.org/10.18653/v1/2026.acl-long.583>
- [43] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2025. tau-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In International Conference on Learning Representations (ICLR). 9965–10017. https://proceedings.iclr.cc/paper_files/paper/2025/hash/1b126cc38b8638e07bef37e7b2bb72bf-Abstract-Conference.html
- [44] Fanghua Ye, Jarana Manotumruksa, and Emine Yilmaz. 2022. MultiWOZ 2.4: A Multi-Domain Task-Oriented Dialogue Dataset with Essential Annotation Corrections to Improve State Tracking Evaluation. In Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue. Association for Computational Linguistics, Edinburgh, UK, 351–360. <https://doi.org/10.18653/v1/2022.sigdial-1.34>
- [45] Steve Young, Milica Gašić, Blaise Thomson, and Jason D. Williams. 2013. POMDP-Based Statistical Spoken Dialog Systems: A Review. *Proc. IEEE* 101, 5 (2013), 1160–1179. <https://doi.org/10.1109/JPROC.2012.2225812>
- [46] Xiaoxue Zang, Abhinav Rastogi, Srinivas Sunkara, Raghav Gupta, Jianguo Zhang, and Jindong Chen. 2020. MultiWOZ 2.2: A Dialogue Dataset with Additional Annotation Corrections and State Tracking Baselines. In Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI. Association for Computational Linguistics, 109–117. <https://doi.org/10.18653/v1/2020.nlp4convai-1.13>
- [47] Guilin Zhang, Wei Jiang, Xiejiaohan Wang, Aisha Behr, Kai Zhao, Jeffrey Friedman, Xu Chu, and Amine Anoun. 2026. Adaptive Memory Admission Control for LLM Agents. <https://doi.org/10.48550/arXiv.2603.04549> arXiv:2603.04549
- [48] Michael JQ Zhang and Eunsol Choi. 2025. Clarify When Necessary: Resolving Ambiguity Through Interaction with LMs. In Findings of the Association for Computational Linguistics: NAACL 2025. Association for Computational Linguistics, 5541–5558. <https://doi.org/10.18653/v1/2025.findings-naacl.306>
- [49] Yan Zhang and Shibo Li. 2026. ConsistencyGate: Preventing Memory Contamination in LLM Agents via SelfConsistency Admission Control. <https://doi.org/10.48550/arXiv.2607.22962> arXiv:2607.22962
- [50] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In Advances in Neural Information Processing Systems, Vol. 36. Curran Associates, Inc., 46595–46623. <https://doi.org/10.52202/075280-2020>
- [51] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, 17 (2024), 19724–19731. <https://doi.org/10.1609/aaai.v38i17.29946>